Definite Integrals and the FTC — Bounds, Sign and Variable-of-Integration Errors

Answer Key

AP Calculus AB/BC

Quick Answers

1. -21	4. -7	7. 78
2. 18, $x^3 - 4x$	5. 0.5	8. —
3. 12	6. -5	

Curriculum

Level: AP Calculus AB/BC
FUN-6.D–FUN-6.E – *problem 7*
LIM-5, FUN-6 – *problems 1, 2, 3, 4, 5, 6, 8*
Difficulty 1–5 of 5 across 9 tagged checks.

Common wrong answers

1. *If they got 21: evaluated from the smaller bound to the larger one, ignoring that the given bounds run backwards*
2. *If they got 10: dropped the parentheses around the value at the lower bound, so only the first piece of it was subtracted*
3. *If they got 14: replaced the variable of integration with the upper bound instead of integrating, so the integrand itself was evaluated*
4. *If they got 7: subtracted the two given integrals but reported the result for the bounds in increasing order, leaving the sign unflipped*
5. *If they got -0.5: used positive cosine as the antiderivative of sine, which reverses the sign of the whole result*
6. *If they got 5: read off the integrand at the moving bound but forgot the minus sign that comes from the variable sitting at the lower bound*
7. *If they got 2: substituted a new variable but kept the original bounds, so the antiderivative in the new variable was evaluated at the old numbers*

1. Find and fix: $\displaystyle\int_5^2 2x\,dx$.

Step 1: name the slip. The FTC says $\int_a^b f = F(b) - F(a)$, with a the *written* lower bound. Here $a = 5$ and $b = 2$, so the subtraction is $F(2) - F(5)$. Jae silently reordered the bounds into increasing order.
Step 2: with $F(x) = x^2$, evaluate in the order the integral was written: $F(2) - F(5) = 4 - 25$.
Step 3: the value is $\boxed{-21}$.
Step 4: check with the reversal rule — $\int_2^5 2x\,dx = 25 - 4 = 21$, and $\int_5^2 = -\int_2^5 = -21$ ✓.
Watch for: 21, the right magnitude with the wrong sign. A backwards integral of a positive function must come out negative.

2. Find and fix: $\displaystyle\int_1^3 (3x^2 - 4)\,dx$.

(a) Step 1: the antiderivative is $F(x) = \boxed{x^3 - 4x}$, since differentiating it returns $3x^2 - 4$.
(b) Step 2: name the slip. Ravi lost the parentheses around $F(1)$. The FTC subtracts the *whole* value $F(1) = 1 - 4 = -3$, so the minus sign must reach both terms: $-(1 - 4) = +3$, not $-1 - 4$.
Step 3: $F(3) - F(1) = (27 - 12) - (1 - 4) = 15 - (-3) = 15 + 3$, so the integral is $\boxed{18}$.
Step 4: check by splitting the integrand: $\int_1^3 3x^2\,dx = 27 - 1 = 26$ and $\int_1^3 4\,dx = 8$, and $26 - 8 = 18$ ✓.
Watch for: 10, exactly 8 low — twice the term that lost its sign.

3. Find and fix: $\displaystyle\int_0^2 (3t^2 + 2)\,dt$.

Step 1: name the slip. Noor substituted the bound into the *integrand*. The bounds are only used after antidifferentiation; the integrand is what gets integrated, never what gets evaluated.
Step 2: antidifferentiate with respect to t: $F(t) = t^3 + 2t$.
Step 3: $F(2) - F(0) = (8 + 4) - 0 = \boxed{12}$.
Step 4: sanity check by size — on $[0, 2]$ the integrand runs from 2 up to 14, so the integral must lie between $2 \times 2 = 4$ and $2 \times 14 = 28$; 12 fits and is nearer the low end, as it should be for a curve that stays small until late ✓.
Watch for: 14. That is the integrand's own value at $t = 2$, which is the tell-tale of this error: the answer equals f(upper bound).

4. Find and fix: $\displaystyle\int_6^4 f(x)\,dx$ **from** $\int_1^6 f = 10$ **and** $\int_1^4 f = 3$.

(a) Step 1: additivity gives $\int_1^4 f + \int_4^6 f = \int_1^6 f$, so $\int_4^6 f = 10 - 3 = 7$. The student's 7 is that integral — correct, but for the bounds in *increasing* order, which is not what was asked.
(b) Step 2: the requested integral runs from 6 down to 4, so it is the negative of the one just found: $\int_6^4 f = -\int_4^6 f = -(10 - 3)$.
Step 3: $\int_6^4 f(x)\,dx = \boxed{-7}$.
Step 4: check by chaining the reversal directly: $\int_6^4 f = \int_6^1 f + \int_1^4 f = -10 + 3 = -7$ ✓.
Watch for: 7. Whenever the answer differs from the right one only in sign, look at the bound order before looking at the arithmetic.

5. Find and fix: $\displaystyle\int_0^{\pi/3} \sin x\,dx$**.**

Step 1: name the slip. The antiderivative of $\sin x$ is $-\cos x$, because $\frac{d}{dx}(-\cos x) = \sin x$ while $\frac{d}{dx}(\cos x) = -\sin x$. Lena's version is the antiderivative of $-\sin x$, so her answer comes out with every sign reversed.
Step 2: evaluate the correct antiderivative: $\big[-\cos x\big]_0^{\pi/3} = -\cos(\pi/3) + \cos(0)$.
Step 3: $= -\dfrac{1}{2} + 1 = \boxed{0.5}$.
Step 4: check by sign reasoning — $\sin x > 0$ on $(0, \pi/3)$, so the integral must be positive. Lena's -0.5 is negative and can be rejected before any arithmetic is examined ✓.
Watch for: -0.5. Always run the positive-integrand test on a trig integral: it catches a flipped antiderivative in one line.

6. Find and fix: $G(x) = \displaystyle\int_x^5 (t^2+1)\,dt$**, find** $G'(2)$**.**

(a) Step 1: FTC Part 1 differentiates an accumulation whose *upper* bound is the variable. Here x is the lower bound, so flip it first: $G(x) = -\displaystyle\int_5^x (t^2+1)\,dt$. The reversal contributes the minus sign Kwame dropped.
Step 2: differentiate the flipped form: $G'(x) = -(x^2+1)$.
(b) Step 3: $G'(2) = -(4+1) = \boxed{-5}$.
Step 4: check by computing G explicitly: $G(x) = \big[\frac{t^3}{3} + t\big]_x^5 = \frac{140}{3} - \frac{x^3}{3} - x$, so $G'(x) = -x^2 - 1$, giving -5 at $x = 2$ ✓. It also makes sense: raising x shrinks the interval $[x, 5]$, so G must decrease.
Watch for: 5. The magnitude is right, so the only thing to check is which bound the variable occupies.

7. Find and fix: $\displaystyle\int_0^2 x(x^2+1)^3\,dx$ **by substitution.**

(a) Step 1: with $u = x^2 + 1$ we get $du = 2x\,dx$, so $x\,dx = \frac{1}{2}du$. The bounds are x-values and must be converted: $x = 0 \Rightarrow u = 1$ and $x = 2 \Rightarrow u = 5$. The new interval is u from 1 to 5.
Step 2: name the slip. Ines evaluated a u-antiderivative at x-bounds — two different variables, one pair of numbers.
(b) Step 3: $\displaystyle\int_1^5 \frac{u^3}{2}\,du = \left[\frac{u^4}{8}\right]_1^5 = \frac{625}{8} - \frac{1}{8} = \frac{624}{8}$, so the integral is $\boxed{78}$.
Step 4: check by expanding instead of substituting: $x(x^2+1)^3 = x^7 + 3x^5 + 3x^3 + x$, whose integral over $[0, 2]$ is $32 + 32 + 12 + 2 = 78$ ✓.
Watch for: 2. Unconverted bounds usually produce an answer far too small, because u-values are shifted away from the x-values that were used.

8. Explain: reversal, dummies, and substitution.

Open response — flagged for manual review. Model answer below; accept any equivalent reasoning.

(a) By the FTC, $\int_a^b f = F(b) - F(a)$ and $\int_b^a f = F(a) - F(b)$. These two expressions are negatives of each other for every F, so swapping the bounds negates the integral. Geometrically, the definite integral is a *signed* area with a direction built in: traversing the interval right-to-left reverses the sign of every dx, so the same region is counted negatively. The magnitude of the area is unchanged; only the orientation is.

(b) In $\int_a^b f(t)\,dt$ the variable t is bound by the integral: it takes every value between a and b and is gone by the time the value is reported. The answer depends on f, a and b only, so writing t in it is meaningless — $\int_0^2 3t^2\,dt$ and $\int_0^2 3s^2\,ds$ are the same number. An accumulation function is different because its *bound* is the variable: in $F(x) = \int_a^x f(t)\,dt$ the letter t is still consumed, but x survives as the input, and FTC Part 1 gives $F'(x) = f(x)$ — the integrand's rule evaluated at the moving bound.

(c) A substitution replaces x by u everywhere, and the bounds are x-values. Leaving them attached to a u-antiderivative evaluates the right function at the wrong two numbers, which is not a small error — it silently answers a different question. Two habits prevent it: write the converted bounds onto the integral sign *at the moment* you write $u = \ldots$, so the integral is never in a half-converted state; or skip conversion entirely by back-substituting to x before evaluating.

What a full-credit answer must contain: the $F(b) - F(a)$ argument (or an equivalent signed-area argument) in (a), an explicit statement that the variable of integration is bound/consumed while the bound variable of an accumulation function is not in (b), and in (c) an account of evaluating a u-expression at x-numbers plus a concrete preventive habit.